

Financial Valuation and Analysis Using R

Individual Assignment

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Introduction

In the contemporary financial landscape, the ability accurately value assets and quantity are a basic skill for financial analyst and data scientists. This report presents a exact computational analysis of various investment instruments within the Sri Lankan market context, Using the R programming language to perform complex financial modeling. As the Sri Lankan economy continues to evolve, the secondary market for debt and equity has become increasingly sophisticated, need investors to move beyond intuition and depend on data-driven terminal valuation model.

The primary objective of this report is to evaluate three distinct financial scenarios. First, we examine bond valuation for Sunrise Holding PLC, focusing on how market yields determined by broader economic condition and Central Bank policies, dictate the present value of fixed income streams. Second, we provide an equity valuation for Oceanic Beverages PLC using a non-constant growth discount model. This model is particularly relevant for companies experiencing a “super growth” phase before settling into a mature, eternal growth cycle. Finally, the report assesses investment performance by comparing historical data from Tech Nova PLC against probabilistic forecasts for Eco Grow PLC.

By applying statistical measures such as mean, variance, and standard deviation, this analysis aims to provide a clear picture of risk-return trade-offs inherited in each investment. The integration of R-based scripts is sure that the valuations are not only accurate but also reproducible, allowing for sensitivity analysis as market variables fluctuate.

Session 01: Bond Valuation Scenario

In the Sri Lankan context, the most traded bonds are the bonds issued by the Central Bank of Sri Lanka on behalf of the government of Sri Lanka. After the issue of the bond, the bond may trade in the secondary market. Then the secondary market comprises with the primary dealers (appointed by the Central Bank of Sri Lanka), licensed commercial banks, institutional investors and individual investors.

The valuation of the bond will depend on the Required Rate of Return (RRR).

$$V_{coupons} = \sum \frac{C}{(1+r)^t}$$

$$V_{face\ value} = \frac{F}{(1+r)^T}$$

Where:

C = Coupon payments, future cash flows.

r = Market interest rate

F = Face value of the bond

T = Time to maturity

t = number of periods

Consider a fixed-income security issued by **Sunrise Holding PLC**. The bond has the following characteristics:

- A nominal (face) value of LKR 5,000
- Annual coupon payments calculated at a rate of 7%
- The market yield currently prevailing at 9%
- A remaining life to maturity of 10 years.

Using the above data:

- i. Annual cash flows to the bondholder in the form of coupon payments.
- ii. The present value of the stream of coupon payments using the appropriate discount rate.
- iii. The present value of the principal repayment is expected at maturity.
- iv. Derive the estimated fair value of the bond based on the sum of the present values.
- v. A brief discussion on how changes in the market rate could influence the valuation of such a bond.

R code

```
##{r}
# Bond Parameters
face_value <- 5000      # Face value of the bond
coupon_rate <- 0.07     # Annual coupon rate (7%)
market_rate <- 0.09     # Market interest rate (9%)
years_to_maturity <- 10 # maturity of 10 years

# Calculate annual coupon payment
coupon_payment <- face_value * coupon_rate

# calculates the present value of all future coupon payments from a bond
pv_coupons <- sum(coupon_payment / (1 + market_rate)^(1:years_to_maturity))

# Calculate present value of face value
pv_principal <- face_value / (1 + market_rate)^years_to_maturity

# Total fair value of bond
fair_value <- pv_coupons + pv_principal

# Output the result
cat("Annual Coupon Payment: LKR", coupon_payment, "\n")
cat("PV of Coupon Stream: LKR", round(pv_coupons, 2), "\n")
cat("PV of Principal: LKR", round(pv_principal, 2), "\n")
cat("Fair Value of the Bond: ", round(fair_value, 2), "\n")
...
```

```
Annual Coupon Payment: LKR 350
PV of Coupon Stream: LKR 2246.18
PV of Principal: LKR 2112.05
Fair Value of the Bond: 4358.23
```

- Annual Coupon Payment: LKR 350
 - Present value of Coupon Stream: LKR 2246.18
 - Present value of Principal: LKR 2112.05
 - Fair value of the Bond: LKR 4358.23
-
- When the Fair value of the bond (which is LKR 4358.23) is less than Face value of the bond (which is LKR 5000, at this instant, the bond is said to trade at a “Discount”. This status arises because the market yield currently prevailing at 9% , it is higher than the annual coupon payments calculated at a rate of 7%. If market interest rates rise further, the value of this bond will decrease. Then market rates will fall, and the bond’s price will increase.

Session 02: Equity Valuation Scenario

Techniques to value equity

01. Technical Analysis

Technical Analysis is the valuation of equity based on historical trends in the market. Based on the technical analysis there are two strategies an investor can take,

Momentum Strategy: A typical investor may buy equities when others buy some equity and when the prices are escalating. Similarly, an investor may sell equities when everyone else is selling their equities and prices are declining.

Contrarian strategy: The goal of a contrarian strategy is to buy equities that are undervalued and sell equities that are overvalued in the market.

02. Fundamental Analysis

In fundamental analysis , a share is valued based on its future economic benefits. In equity valuation, we use assumptions and subsequently separate equation to value equity.

1) Assumption 01: One period dividend discount model

Assume that an investor will hold onto equity only for one year.

$$P_o = \frac{D_1 + P_1}{(1+r)}$$

Po= Fair value of an equity

D1= Dividends expected at the end of year 01

P1= Selling price of the equity after one year

r = Required rate of return

2) Assumption 02: Multi period dividend discount model

I. Scenario 01: Zero growth in dividends.

Assume that an investor holds on equity for relatively long time and the dividends of the stated equity will remain the same.

$$P_o = \frac{D}{r}$$

Po = Fair value of an equity

D= Dividend received

r = Required rate of return

II. Scenario 02: Constant growth in dividends

This scenario assumes that the firm pays the dividends every year and its growth is at a given constant percentage yearly.

$$P_0 = \frac{D_0(1+g)}{(r-g)}$$

P_0 = Fair value of the equity

D_0 = Dividends paid in the current year

g = Dividends growth rate

r = Required rate of return

III. Scenario 03: Non- constant growth in dividends.

A first stage with a high growth rate for n years, the ‘supernatural’ period.

A second stage with a constant growth rate forever, the ‘normal’ period.

$$P_0 = \sum_{t=1}^n \frac{D_0(1+g)^t}{(1+r)^t} + \frac{P_n}{(1+r)^n}$$

$$P_n = \frac{D_{n+1}}{r-g_L}$$

P_0 = Current stock price

D_0 = Most recent dividend

P_n = Stock price at year n

D_n = Dividend in year n

D_{n+1} = First dividend of normal growth stage

g = High growth rate

g_L = Long-run constant growth rate

r = Required rate of return

t = time period

For Oceanic Beverages PLC, we employ a non-constant growth discount model.

An investor is analyzing shares of **Oceanic Beverages PLC** and has gathered the following information:

- The most recent dividend paid (D_0) was LKR 3.50 per share.
- The company expects dividends to grow at a rate of 12% per annum for the next four years.
- Beyond year four, dividends are anticipated to grow at a perpetual rate of 5%.
- The investor requires a return of 14% to justify the investment, considering market conditions and company-specific risk factors.

Using the above data:

- i. Project the anticipated dividends for the next four years.
- ii. Estimate the expected dividend in the fifth year based on perpetual growth.
- iii. Calculate the terminal value of the equity at the end of year four using the constant growth model.
- iv. Discount each projected dividend and the terminal value to present value.
- v. Arrive at the fair value per share of Oceanic Beverages PLC under this multi-stage dividend discount framework.
- vi. How sensitive the valuation is to changes in the perpetual growth rate and the investor's required rate of return.

R code

```
##{r}
# Scenario 03: Non-Constant growth in dividends
# Inputs
D0 <- 3.5          # Last dividend paid
g1 <- 0.12         # Growth rate for first 4 years
g2 <- 0.05         # Growth rate thereafter
r  <- 0.14         # Required rate of return

# Step 1: Calculate dividends for years 1 to 4
D1 <- D0 * (1 + g1)
D2 <- D1 * (1 + g1)
D3 <- D2 * (1 + g1)
D4 <- D3 * (1 + g1)

# Step 2: Calculate terminal value at end of year 4
TV <- D4 * (1 + g2) / (r - g2)

# Step 3: Discount dividends and terminal value to present
PV_D1 <- D1 / (1 + r)^1
PV_D2 <- D2 / (1 + r)^2
PV_D3 <- D3 / (1 + r)^3
PV_D4 <- D4 / (1 + r)^4
PV_TV <- TV / (1 + r)^4

# Step 4: Calculate total present value (share price)
fair_value <- PV_D1 + PV_D2 + PV_D3 + PV_D4 + PV_TV

# Output result
cat("Projected D1-D4:", round(D1,2), round(D2,2), round(D3,2), round(D4,2), "\n")
cat("Terminal Value at Year 4: LKR", round(TV, 2), "\n")
cat("Fair Value per Share: LKR", round(fair_value, 2), "\n")
##}
```

```
Projected D1-D4: 3.92 4.39 4.92 5.51
Terminal Value at Year 4: LKR 64.25
Fair Value per Share: LKR 51.44
```

- Projected anticipated values:

D1: 3.92

D2: 4.39

D3: 4.92

D5: 5.51

- Terminal Value at Year 4: LKR 64.25
- Fair value per share: LKR 51.44
- The valuation is highly sensitive to the perpetual growth rate and the required rate of return. Because the terminal value constitutes most of the share price, a small increase in the required return or a decrease in long-term growth would lead to a substantial drop in the fair value per share.

Session 03: Analysis of Historical Investment Performance

The expected return and the anticipated risk of an investment can be determined through,

01. Historical data

02. Forecasted data

Expected Return and the Risk of an investment based on historical data.

$$\text{Expected Return (ER): } \frac{\sum R_{(i)}}{N}$$

$R_{(i)}$ = return in period i

N= Total number of periods

$$\text{Risk of an investment: } \sigma = \sqrt{\frac{\sum_{i=1}^n (r_i - \bar{r})^2}{n-1}}$$

σ = Standard deviation

r_i = Return in period i

$\bar{r}$ = Average return

n = number of observations

The following annual returns were made over six consecutive years for **Tech Nova PLC**:

Year	Annual Return ($R_{(i)}$)
1	18%
2	10%
3	4%
4	12%
5	20%
6	8%

Based on this historical performance:

- i. Determine the mean annual return.
- ii. Assess the variability of returns by calculating the standard deviation.
- iii. Discuss how these measures provide insights into the investment's risk profile.

R code

```
##{r}
# Annual returns in decimal form
returns <- c(0.18, 0.10, 0.04, 0.12, 0.20, 0.08)

# Calculate average return
average_return <- mean(returns)

# Calculate sample standard deviation (risk)
std_deviation <- sd(returns)

# Display results in percentage format
cat("Average Return:", round(average_return * 100, 2), "%\n")
cat("Standard Deviation (Risk):", round(std_deviation * 100, 2), "%\n")
##}
```

```
Average Return: 12 %
Standard Deviation (Risk): 6.07 %
```

- Average return : 12%
- Risk of an investment (Standard Deviation): 6.07%
- These measures quantify the trade-off between reward and uncertainty. A 12% mean return coupled with a 6.07% standard deviation indicates moderate volatility; The actual return has historically deviated from the average by roughly 6% in either direction.

Expected Return and the Risk of an investment based on forecasted data.

Expected return (ER): $\sum_{i=1}^n (r_i \times prob_i)$

Risk of an Investment: $\sigma = \sum_{i=1}^n (prob_i) \times (r_i - ER_i)^2$

r_i = Return in scenario i

$prob_i$ = Probability of scenario i

n = Total number of possible scenarios

An investment analyst is reviewing potential future performance scenarios for **Eco Grow PLC**. The company's management projects three possible return outcomes, along with the likelihood of each scenario occurring:

Possible Annual Return (%)	probability
25	0.25
12	0.50
5	0.25

Given these projections:

- Compute the expected (mean) return for Eco Grow PLC.
- Estimate the variance and standard deviation of the potential returns.
- Provide a brief commentary on how the distribution of possible returns influences the perceived risk of the investment.

R code

```

## {r}
# Define returns and probabilities
returns_f <- c(0.25, 0.12, 0.05)      # Convert % to decimals
probabilities <- c(0.25, 0.50, 0.25)  # Probabilities must sum to 1

# Step 1: Expected return
expected_return <- sum(returns_f * probabilities)

# Step 2: Variance
variance <- sum(probabilities * (returns_f - expected_return)^2)

# Step 3: Standard deviation
std_deviation <- sqrt(variance)

# Output results
cat("Expected Return:", round(expected_return * 100, 4), "%\n")
cat("Variance:          ", round(variance * 10000, 2), "\n")
cat("Standard Deviation (Risk):", round(std_deviation * 100, 4), "%\n")

```

```

Expected Return: 13.5 %
Variance:          52.25
Standard Deviation (Risk): 7.2284 %

```

- Expected Return: 13.5%
- Variance: 52.25
- Risk of investment (Standard Deviation): 7.2284%
- The distribution of returns influences the perceived risk. While Eco Grow PLC offers a higher expected return(13.5%) than Tech Nova PLC (12%), it also presents higher risk as evidences by its standard deviation (7.23% vs 6.07%). The 25% probability of achieving only a 5% return highlights the potential for significant downside compared to the expected mean.

Portfolio Theory

This theory explains how investors can minimize risk and maximize return by diversifying their investment across a range of investment opportunities.

Coefficient of variation = Risk per unit of return

$$= \frac{\text{Standard deviation}}{\text{Expected return}}$$

	Tech Nova PLC	Eco Grow PLC
Expected Return	12%	13.5%
Standard Deviation (Risk)	6.07%	7.2284%
Coefficient of variation	$\frac{0.0607}{0.12} = 0.5058$	$\frac{0.072284}{0.135} = 0.5354$
Which investment to select	Select	

Conclusion

This comprehensive analysis has demonstrated that financial valuation is a dynamic process where small changes in underlying assumptions can lead to major different investment conclusions.

In session 01, our analysis of the Sunrise Holding PLC bond revealed a fair value of LKR 4,358.23, which is significantly lower than its face value. The core conclusion here is that in a high-interest rate environment where market yields(9%) exceed coupon rates(7%), debt instrument will inevitably trade at a discount. For Sunrise Holding, this discount reflects the market's requirement for a higher return to offset the opportunity cost of capital.

In session 02, the valuation of Oceanic Beverages PLC at LKR 51.44 per share underscores the importance of the terminal value in equity modeling. Because the terminal value accounts for the majority of the share's worth, the conclusion for investors must be one of caution, the valuation highly sensitive to the perpetual growth rate and the required return. A slight decrease in long-term growth expectations could render the stock overvalued at its current price.

In session 03, The comparison between Tech Nova and Eco Grow PLC highlights the fundamental rule of finance; there is no return without risk. While Eco Grow PLC offers higher expected return of 13.5%, it carries a higher standard deviation of 7.23%. The 25% probability of a mere 5% return for Eco Grow represents a significant downside risk that conservative investors must weigh against the 12% stable return of Tech Nova.

Overall, the R-based methodologies employed throughout this report provide a robust framework for objective decision making. By quantifying the relationship

between reward and uncertainty, data science empowers investors to navigate the complexities of the Sri Lankan financial markets with greater precision and confidence.